GROUP WEEKLY REPORTS

Project Title: Why 2024 Flights Ran Late
Team: Team 10 – DATA-VIZ HACK-A-BIT
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WEEK 7 – FEATURE ENGINEERING AND DATA TRANSFORMATION

During the seventh week, our team focused on feature engineering and data transformation to improve the suitability of the dataset for statistical analysis and potential predictive modeling.

We examined the existing variables and discussed how additional features could be derived from the available information. The team explored the importance of preparing numerical and categorical variables in appropriate formats for machine learning algorithms.

We worked with categorical encoding techniques to convert non-numerical variables into suitable numerical representations. We also explored scaling techniques to bring numerical variables

to comparable scales and reduce the influence of differences in magnitude.

Since some numerical variables exhibited skewed distributions, we examined the Yeo-Johnson transformation as a method for reducing skewness and improving the distribution of data. We discussed how this transformation differs from other techniques and why it can be applied to values that include zeros and negative numbers.

The team also explored the differences between normalization and standardization and understood their importance in data preprocessing.

We reviewed the transformed variables and discussed how these operations could influence subsequent analysis. We ensured that preprocessing decisions were appropriate for the intended analytical objectives.

By the end of the week, we had gained a better understanding of feature engineering, encoding, scaling, and data transformation. These activities helped prepare the dataset for correlation analysis and dimensionality reduction.